

OptiCrop – Smart Agricultural Production Optimization Engine

Project Description:

The project “Smart Agricultural Production Optimization Engine” aims to develop an advanced software system that utilizes data-driven insights to optimize agricultural production for different crops. By integrating key environmental factors such as Nitrogen (N), Phosphorous (P), Potassium (K) levels, soil temperature, humidity, pH, rainfall, and crop types, OptiCrop seeks to provide intelligent recommendations to farmers for maximizing yields and resource efficiency. The knowledge and insights gained through the OptiCrop project can be utilized by agricultural researchers, policymakers, and stakeholders to advance the understanding of crop-environment interactions and inform the development of evidence-based agricultural strategies. The OptiCrop project revolutionizes agricultural practices by providing farmers with a smart production optimization engine. By integrating key environmental factors and crop-specific recommendations, OptiCrop enables farmers to achieve optimal yields, improve resource efficiency, and contribute to sustainable agricultural practices.

Scenarios

Scenario 1: Smart Crop Recommendation for Farmers

A farmer enters soil and environmental details such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH level, and rainfall. The system analyzes the data and recommends the most suitable crop for maximum yield and better farming efficiency.

Scenario 2: Crop Suitability and Environmental Assessment

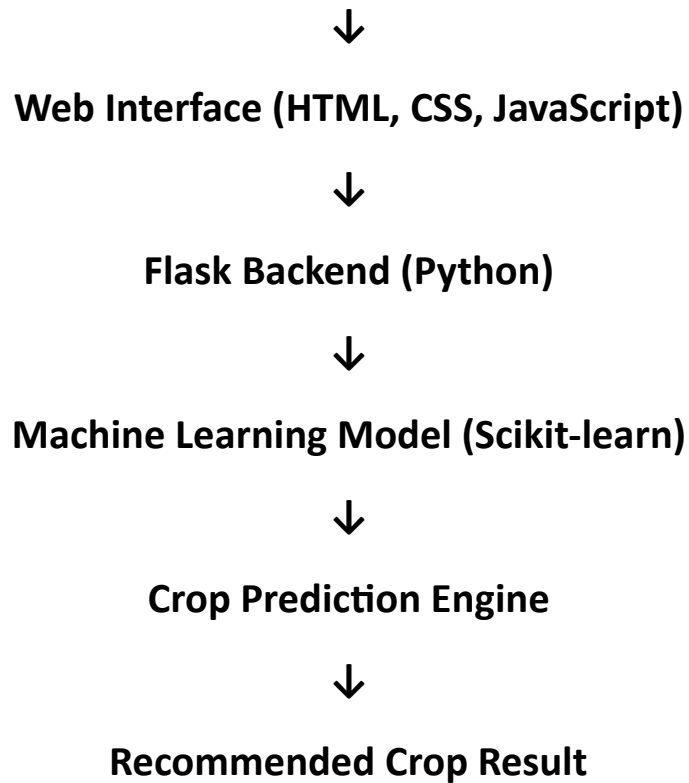
A user wants to understand whether the current soil and climate conditions are suitable for a particular crop. The application evaluates the input parameters and provides insights about crop compatibility, environmental suitability, and productivity potential.

Scenario 3: Agricultural Research and Policy Planning

An agricultural researcher or policymaker uses the system to analyze crop-environment relationships and identify patterns in agricultural production. The platform helps in making data-driven decisions for sustainable farming strategies and resource optimization.

Technical Architecture:

User



Description:

The OptiCrop system follows a three-layer architecture consisting of the User Interface Layer, Application Layer, and Machine Learning Layer. The frontend is developed using HTML, CSS, and JavaScript, allowing users to enter agricultural parameters such as Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall. The Flask backend processes the user input, validates the data, and sends it to the trained Machine Learning model.

The Machine Learning model, developed using Python and Scikit-learn, analyzes the input parameters and predicts the most suitable crop for cultivation. The prediction result is then returned to the Flask application and displayed to the user through the web interface. This architecture provides an efficient, user-friendly, and intelligent crop recommendation system that assists farmers in making data-driven agricultural decisions and improving crop productivity.

Pre-requisites:

- Python

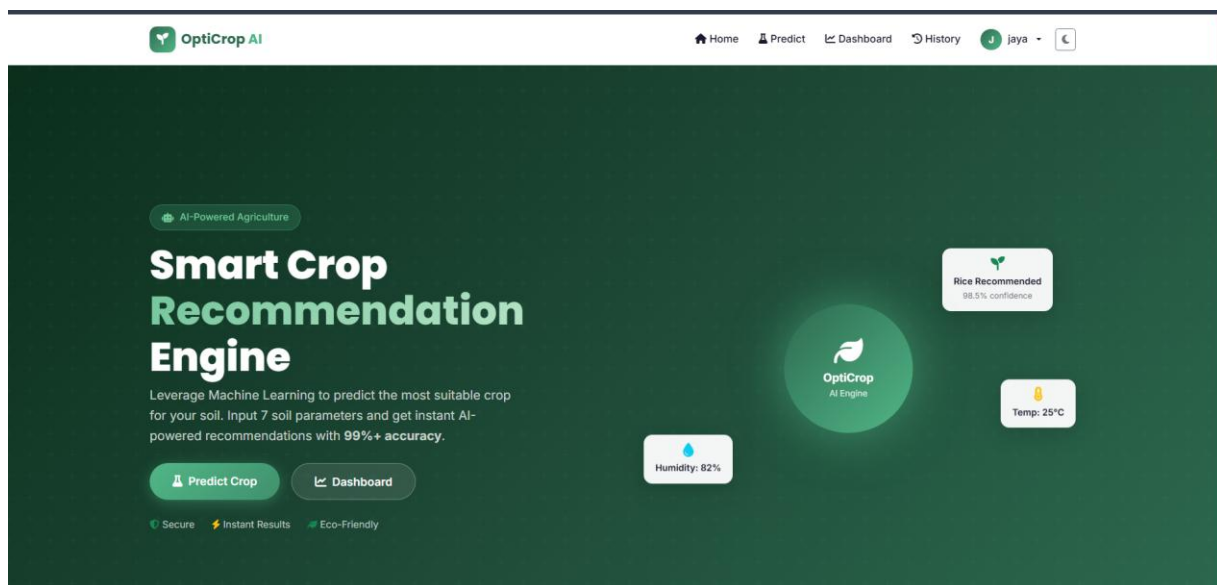
- Flask
- Pandas
- NumPy
- Scikit-learn
- HTML
- CSS
- JavaScript
- Git and GitHub

Project Workflow:

Epic 1: Define Problem and Understanding

Story 1: Identify and define the agricultural problem that the crop recommendation system aims to solve.

Story 2: Gather and analyze business requirements to understand project objectives and expected outcomes.



Story 3: Conduct a literature survey to study existing crop recommendation techniques and machine learning approaches.

Story 4: Analyze the social and business impact of implementing an AI-driven crop recommendation solution.

An AI-powered crop recommendation engine built for SmartBridge Internship using Machine Learning, Flask, and modern web technologies.

Random Forest

Microsoft Excel interface showing the 'Crop_recommendation' dataset. The ribbon includes Home, Insert, Draw, Page Layout, Formulas, Data, Review, View, Help, and Acrobat. The 'Crop_recommendation' workbook is open, displaying a table with columns A through U. The data includes crop names (rice, wheat, sugarcane, etc.) and various attributes (temp, humidity, ph, rainfall, label). The 'Crop_recommendation' tab is selected at the bottom.

Story 4: Perform univariate analysis to examine the distribution and characteristics of individual features.

Story 5: Conduct bivariate analysis to identify relationships between agricultural parameters and crop suitability.

Story 6: Perform multivariate analysis to discover patterns and interactions among multiple variables.

Epic 3: Data Pre-Processing

Story 1: Check for null values and handle missing data to improve dataset quality.

Story 2: Detect and treat outliers that may affect model performance.

The screenshot shows the OptiCrop AI web application interface. At the top, there's a navigation bar with the logo, a user profile 'J. jaya', and links for Home, Predict, Dashboard, and History. Below this is a section titled 'AI PREDICTION' with a sub-header 'Crop Recommendation'. A prompt asks users to enter soil parameters for an AI-powered recommendation. The main form, titled 'Soil Analysis Form', contains three sections: 'Soil Nutrients (mg/kg)' with input fields for Nitrogen (N), Phosphorous (P), and Potassium (K); 'Climate Conditions' with input fields for Temperature (°C) and Humidity (%); and 'Soil & Rainfall' with input fields for Soil pH and Rainfall (mm). Each input field has a typical range displayed below it. At the bottom of the form are two buttons: 'Predict Best Crop' and 'Reset'.

OptiCrop AI

Home Predict Dashboard History J. jaya

AI PREDICTION

Crop Recommendation

Enter your soil parameters below to get an AI-powered crop recommendation

Soil Analysis Form

All 7 parameters are required

Soil Nutrients (mg/kg)

Nitrogen (N)	Phosphorous (P)	Potassium (K)
0 - 140	5 - 145	5 - 205
Typical: 0-140 mg/kg	Typical: 5-145 mg/kg	Typical: 5-205 mg/kg

Climate Conditions

Temperature (°C)	Humidity (%)
8 - 44	14 - 100
Typical: 8-44°C	Typical: 14-100%

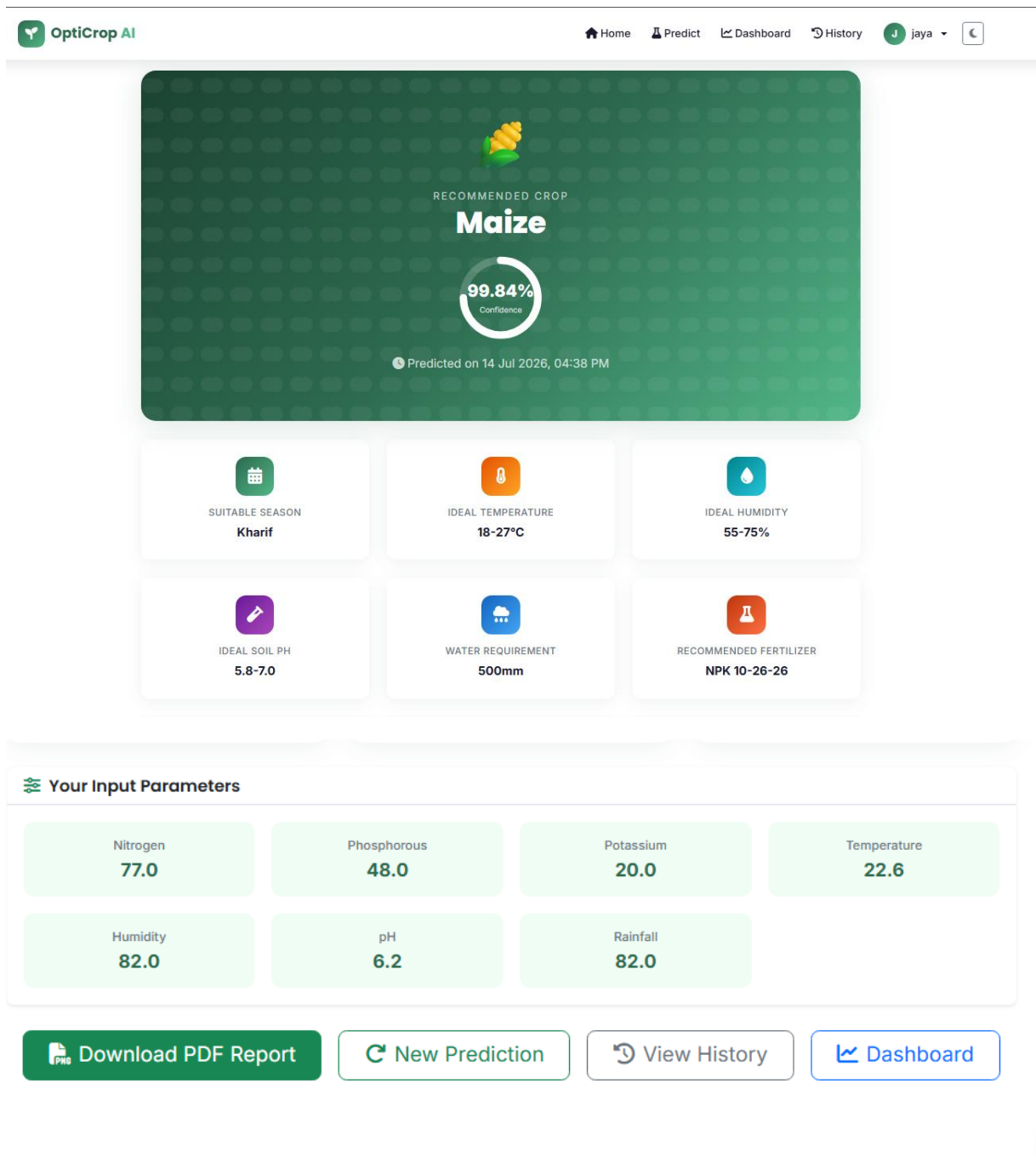
Soil & Rainfall

Soil pH	Rainfall (mm)
3.5 - 9.9	20 - 300
Typical: 3.5-9.9	Typical: 20-300 mm

Predict Best Crop Reset

Story 3: Extract seasonal crop information and prepare relevant features for analysis.

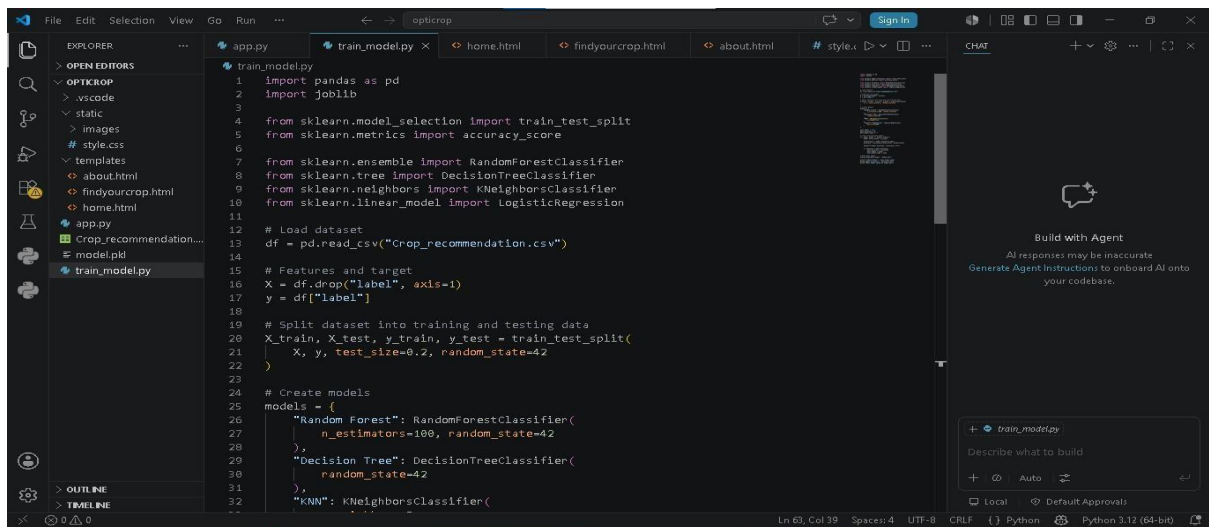
Story 4: Split the dataset into training and testing sets for model development and evaluation.



Epic 4: Model Building

Story 1: Apply K-Means Clustering to identify patterns and group similar agricultural conditions.

Story 2: Train a Logistic Regression model to predict suitable crops based on environmental parameters.



The screenshot shows a Visual Studio Code editor with a dark theme. The Explorer sidebar on the left shows a project structure for 'OPTIKROP' with files like 'style.css', 'home.html', and 'train_model.py'. The main editor window displays the 'train_model.py' file, which contains Python code for loading a dataset, splitting it into training and testing sets, and creating three machine learning models: RandomForestClassifier, DecisionTreeClassifier, and KNeighborsClassifier. The code uses libraries like pandas, joblib, and sklearn. A Chat sidebar on the right is visible, showing a 'Build with Agent' section.

```
1 import pandas as pd
2 import joblib
3
4 from sklearn.model_selection import train_test_split
5 from sklearn.metrics import accuracy_score
6
7 from sklearn.ensemble import RandomForestClassifier
8 from sklearn.tree import DecisionTreeClassifier
9 from sklearn.neighbors import KNeighborsClassifier
10 from sklearn.linear_model import LogisticRegression
11
12 # Load dataset
13 df = pd.read_csv("Crop_recommendation.csv")
14
15 # Features and target
16 X = df.drop("label", axis=1)
17 y = df["label"]
18
19 # Split dataset into training and testing data
20 X_train, X_test, y_train, y_test = train_test_split(
21     X, y, test_size=0.2, random_state=42
22 )
23
24 # Create models
25 models = {
26     "Random Forest": RandomForestClassifier(
27         n_estimators=100, random_state=42
28     ),
29     "Decision Tree": DecisionTreeClassifier(
30         random_state=42
31     ),
32     "KNN": KNeighborsClassifier(
33         n_neighbors=5,
34         metric='minkowski',
35         p=2,
36         weights='uniform',
37         leaf_weight='uniform',
38         algorithm='brute'
39     )
40 }
```

Story 3: Evaluate model performance using appropriate metrics and compare results.

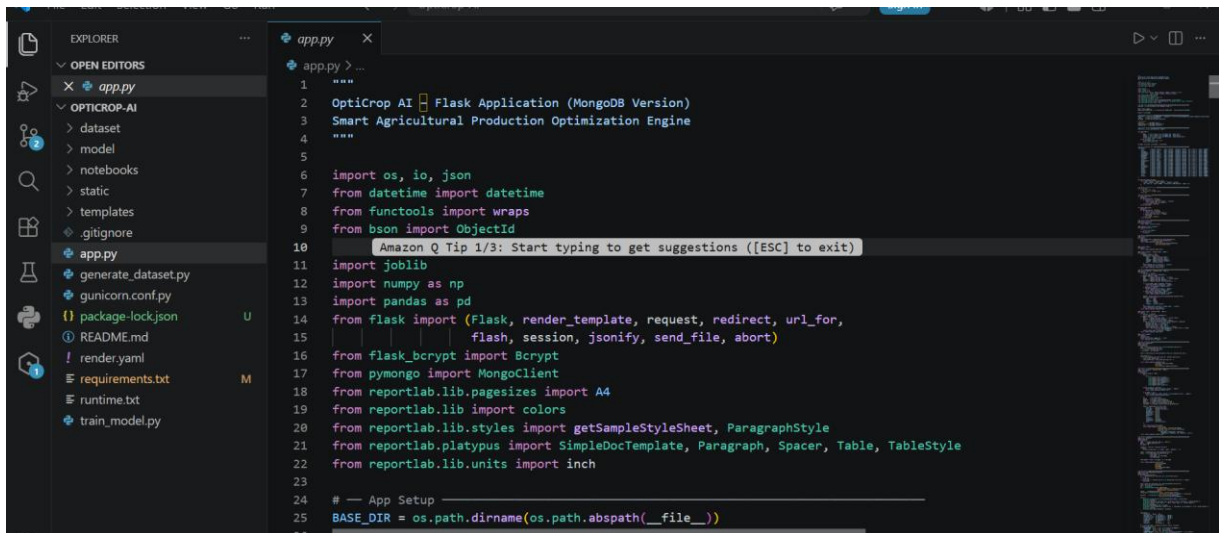
Story 4: Save the best-performing model for deployment and generate crop predictions based on user-provided inputs.

Epic 5: Application Building

Story 1: Design and develop HTML pages to create an interactive and userfriendly interface.

Story 2: Build the Python backend code and integrate it with the trained machine learning model.

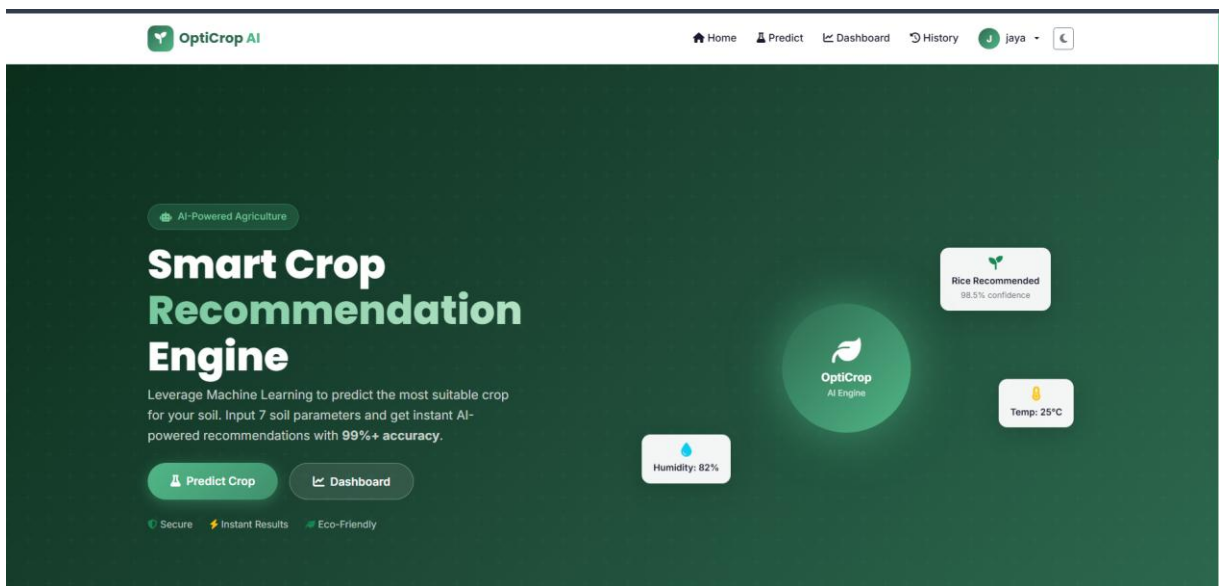
Story 3: Run, test, and validate the application to ensure accurate crop recommendations and smooth functionality.



```
1  """
2  OptiCrop AI Flask Application (MongoDB Version)
3  Smart Agricultural Production Optimization Engine
4  """
5
6  import os, io, json
7  from datetime import datetime
8  from functools import wraps
9  from bson import ObjectId
10
11  Amazon Q Tip 1/3: Start typing to get suggestions ([ESC] to exit)
12
13  import joblib
14  import numpy as np
15  import pandas as pd
16
17  from flask import (Flask, render_template, request, redirect, url_for,
18                    flash, session, jsonify, send_file, abort)
19  from flask_bcrypt import Bcrypt
20  from pymongo import MongoClient
21  from reportlab.lib.pagesizes import A4
22  from reportlab.lib import colors
23  from reportlab.lib.styles import getSampleStyleSheet, ParagraphStyle
24  from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Table, TableStyle
25  from reportlab.lib.units import inch
26
27  # --- App Setup ---
28  BASE_DIR = os.path.dirname(os.path.abspath(__file__))
```

Exploring the Website's Web Pages:

Home Page:



Description:

The Home Page is the main entry point of the OptiCrop – Smart Agricultural Production Optimization Engine. It provides an attractive and user-friendly interface with an agricultural background image and introduces the project to users. The page contains two main buttons: Find Your Crop and About Project. The "Find Your Crop" button redirects users to the crop prediction page, while the "About Project" button provides information about the project

and its objectives. The home page is designed to give users easy navigation and a professional look.

Input Section:

The screenshot shows the 'Crop Recommendation' interface of the OptiCrop AI application. At the top, there's a navigation bar with 'Home', 'Predict', 'Dashboard', and 'History' links, along with a user profile 'jaya'. Below this is a header section with the title 'Crop Recommendation' and a subtitle 'Enter your soil parameters below to get an AI-powered crop recommendation'. The main form is titled 'Soil Analysis Form' and includes a note 'All 7 parameters are required'. The form is divided into three sections: 'Soil Nutrients (mg/kg)' with input fields for Nitrogen (N), Phosphorous (P), and Potassium (K); 'Climate Conditions' with input fields for Temperature (°C) and Humidity (%); and 'Soil & Rainfall' with input fields for Soil pH and Rainfall (mm). Each input field has a range and a typical value. At the bottom, there are two buttons: 'Predict Best Crop' and 'Reset'.

OptiCrop AI Home Predict Dashboard History jaya

AI PREDICTION

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Soil & Rainfall

Soil pH	Rainfall (mm)
3.5 - 9.9	20 - 300
Typical: 3.5-9.9	Typical: 20-300 mm

Predict Best Crop **Reset**

Description:

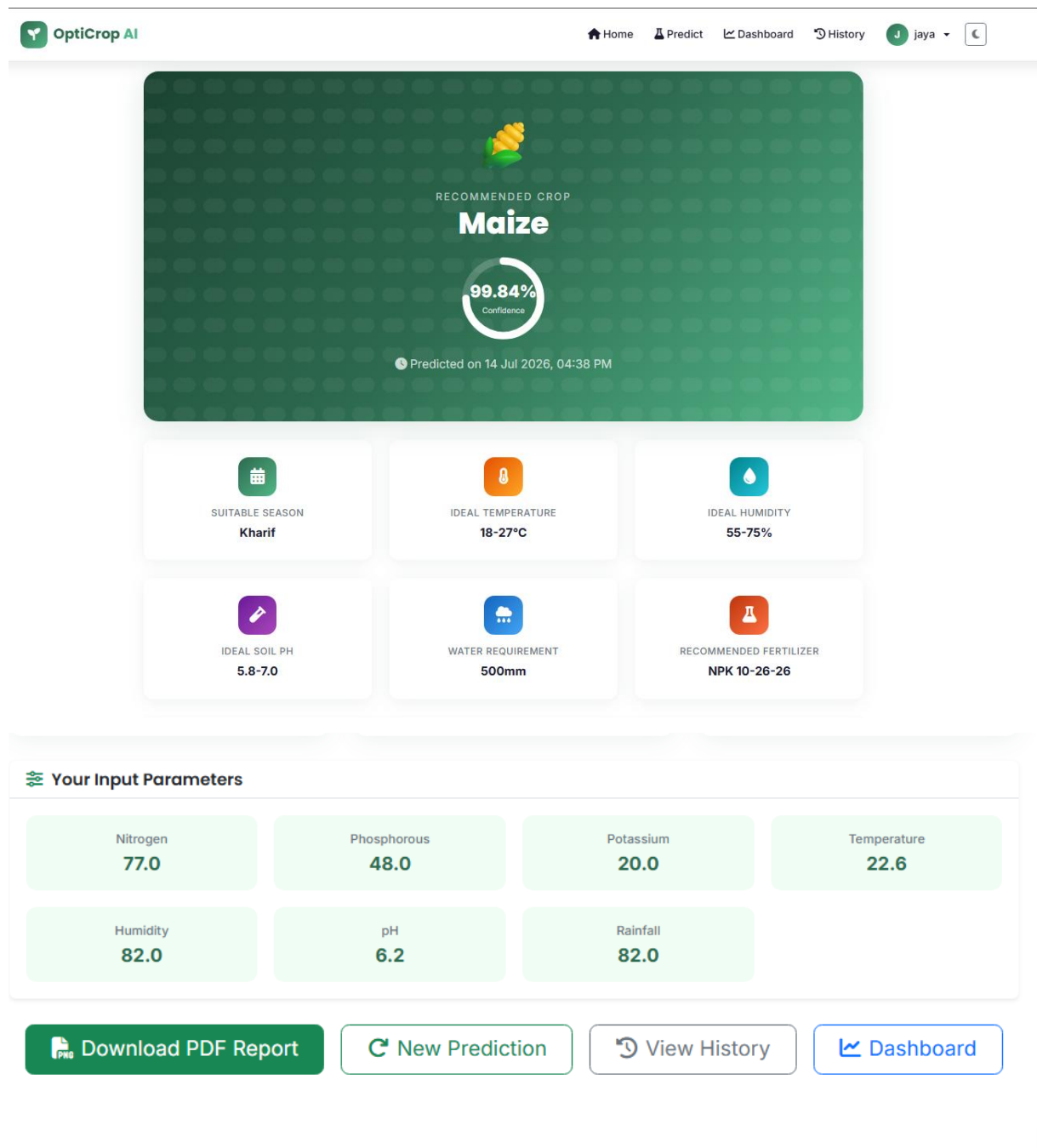
The Find Your Crop page is the core functionality of the project. This page allows users to enter important agricultural parameters such as:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Temperature
- Humidity
- pH Value

- Rainfall

After entering these values, the user clicks the Predict Crop button. The Flask backend sends the data to the trained Machine Learning model, which analyzes the input and predicts the most suitable crop for cultivation. The predicted crop is then displayed to the user. A Clear button is also provided to reset all input fields.

Result Section:



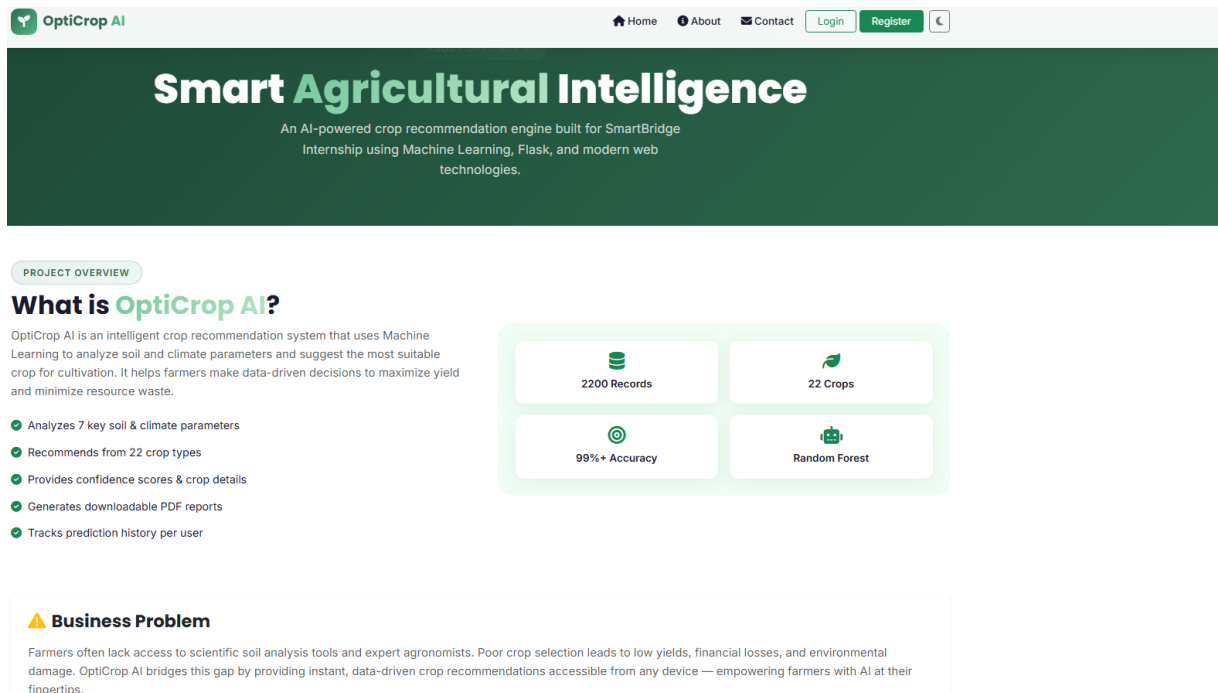
Description:

The OptiCrop – Smart Agricultural Production Optimization Engine was successfully developed and tested. The system accepts agricultural input parameters such as Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall and predicts the most suitable crop using a trained Machine Learning model.

The web application provides accurate and quick crop recommendations through a simple and user-friendly interface. The prediction results help farmers make informed agricultural decisions, improve crop productivity, and select crops that are suitable for their soil and environmental conditions.

The project successfully demonstrates the practical application of Machine Learning in Agriculture by integrating a Flask web application with a crop prediction model.

About Page:



Description:

The About Project page provides detailed information about the purpose and working of the AI-ML OptiCrop system. It explains that the project uses Machine Learning techniques to recommend the most suitable crop based on soil nutrients and environmental conditions. The page describes the input parameters used by the system, including Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall. It also highlights the benefits of the system, such as helping farmers make better agricultural decisions and improving crop productivity.

Conclusion:

OptiCrop is an advanced Machine Learning-powered smart agricultural recommendation system designed to support modern, data-driven farming practices by providing accurate and intelligent crop prediction solutions. The system analyzes multiple critical environmental and soil-related parameters including nitrogen, phosphorus, and potassium (NPK) soil nutrient levels, temperature, humidity, soil pH, and rainfall patterns to determine the most suitable crop for cultivation under specific conditions. By combining agricultural science with artificial intelligence, OptiCrop aims to reduce uncertainty in farming decisions and promote efficient, sustainable agricultural production.

The project leverages multiple Machine Learning algorithms such as K-Nearest Neighbors (KNN), K-Means Clustering, and Logistic Regression to perform crop classification and recommendation tasks. Each algorithm contributes uniquely to the prediction process: KNN identifies similar environmental conditions

based on historical agricultural data, K-Means helps in grouping and analyzing soil and climate patterns, while Logistic Regression provides accurate classification of the most appropriate crop category. The use of multiple models allows comparative analysis and improves the reliability and robustness of predictions, ensuring better recommendation accuracy across varying agricultural conditions.